Question

There is an empirical rule that can be used to estimate the activation energy of elementary reactions. Under the following conditions, the activation energy can be calculated using the corresponding methods in the table below:

① Reactions without bond breaking;

② Exothermic reactions between radicals and molecules;

③ Reactions with incomplete bond breaking (e.g., concerted reactions);

④ Reactions with complete bond breaking.

Activation Energy/BE	0	0.05	0.3	1
	①	②	③	④

BE is the sum of the bond energies of the broken bonds in the reaction.

Based on the above empirical rule and the provided bond energy information, estimate the activation energies of each elementary reaction in the reaction between hydrogen and chlorine, and compare them with the given experimental values.

$(A)\ Cl_2 = 2Cl\cdot\ \ E_{a1} = 242.8 kJ \cdot mol^{-1}$

$(B)\ Cl \cdot + H_2 = HCl + H\cdot$

$E_{a2} = 25.1 kJ \cdot mol^{-1}$

$(C)\ H \cdot + Cl_2 = HCl + Cl\cdot$

$E_{a3} = 12.6 kJ \cdot mol^{-1}$

$BE(Cl-Cl) = 243 kJ \cdot mol^{-1}, BE(H-H) = 436 kJ \cdot mol^{-1}, BE(H-Cl) = 431 kJ \cdot mol^{-1}$

A. The estimated activation energy for reaction (C) is in complete agreement with the experimental value.

B. The activation energy of reaction (A) is 0

C. The estimated activation energy for reaction (B) is $3.3kJ \cdot mol^{-1}$ lower than the experimental value.

D. The estimated activation energy for reaction (B) is greater than the experimental value by $1.4kJ \cdot mol^{-1}$.

E. Using the method described in the problem, the relative error in the activation energy of reaction (C) is calculated to be 3.17%.

F. None of the above options are correct

Answer

Correct Answer: D

Detailed Explanation

Reaction (A): $E_{a1} = BE(Cl - Cl) = 243kJ \cdot mol^{-1}$

CHECKPOINT 1 PTS

$E_{a1} = 243kJ \cdot mol^{-1}$

Reaction (B): $\Delta H = BE(H - H) - BE(H - Cl) = 5kJ \cdot mol^{-1} > 0$

CHECKPOINT 1 PTS

$\Delta H > 0$

The empirical formula $E_a = 0.05 \cdot BE(H - H) = 21.8kJ \cdot mol^{-1}$ cannot be directly applied. Instead, calculate using the relationship: activation energy of the forward reaction - activation energy of the reverse reaction = ΔH:

$$E_{a2} = E_{a2,\text{reverse}} + \Delta H = 0.05 \cdot BE(H - Cl) + 5kJ \cdot mol^{-1} = 26.5kJ \cdot mol^{-1}$$

CHECKPOINT 1 PTS

Calculate using activation energy of forward reaction - activation energy of reverse reaction = ΔH

CHECKPOINT 1 PTS

$E_{a2} = 26.5kJ \cdot mol^{-1}$

Reaction (C): $\Delta H = BE(Cl-Cl) - BE(H-Cl) = -188kJ \cdot mol^{-1} < 0$. The empirical formula $E_{a3} = 0.05 \cdot BE(Cl-Cl) = 12.2kJ \cdot mol^{-1}$ is directly applied.

CHECKPOINT 1 PTS

$E_{a3} = 12.2kJ \cdot mol^{-1}$

Note: The relative error mentioned in option E is -3.17%.